

# Fethi Bencherki, Licentiate

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## Education

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- 2021-Present    **Ph.D. in Automatic Control**, Lund University, Sweden  
Advisor: [Prof. Anders Rantzer](#)  
(GPA: pass/fail only)
- 2021-24        **Licentiate in Automatic Control**, Lund University, Sweden [\[Diploma\]](#)  
(GPA: pass/fail only)  
Thesis title: *Adaptive Control of Positive Systems* [\[link\]](#)
- 2017-20        **M.Sc. in Control Systems**, Eskisehir Technical University, Turkey [\[Diploma\]](#)  
(GPA: 3.96/4.0, Valedictorian)  
Advisor: [Prof. Hüseyin Akçay](#)  
Thesis title: *State-space identification of switched linear systems via sparse optimization* [\[link\]](#)
- 2016-17        **M.Sc. in Control Engineering**, Institute of Electrical and Electronic Engineering,  
Boumerdes, Algeria  
(GPA: 16.69/20, Valedictorian)
- 2012-15        **B.Sc. in Electrical and Electronic Engineering**, Institute of Electrical and Electronic Engineering,  
Boumerdes, Algeria [\[Diploma\]](#)  
(GPA: 15.90/20, Valedictorian)

## Research interests

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My research focuses on learning and control in large-scale networks, with particular emphasis on positive systems. I am also broadly interested in learning-based control and system identification, with a specific focus on hybrid systems.

## Research experience

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- 2021-Present    **Ph.D. student**, [Department of Automatic Control, Lund University](#)
- Mar 2025-Jun 25    **Visiting researcher**, at [Electrical and Systems Engineering Department, University of Pennsylvania](#), (Host: [Prof. Nik Matni](#))

## Industry experience & study trip

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- 2023              Participant in [WASP Study Trip to Boston](#), MA, United States  
Visiting Boston Dynamics, MathWorks, PTC, Motionl, MERL, Robotics lab at Harvard and MIT

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|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jun 2018 - Aug 18 | Trainee at <a href="#">Turkish Airlines, Flight Simulation Department</a> , Istanbul, Turkey<br>Worked on visualizing different flight control processes using an Airbus A330 and Boeing flight simulators, understanding and mastering the simulators operations, fault detection and security procedures |
| Mar 2016 - May 16 | Trainee at <a href="#">Sonelgaz</a> , Chlef, Algeria<br>Understanding the operation of an electricity distribution network within a thermal power station, building electronic circuits from design to final kits mounted on printed circuit boards (PCBs) based on through-hole technology                |

## Awards & honors

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|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025      | <a href="#">Recipient of Kungliga Fysiografiska Sällskapet i Lund, yngre Forskare Doktorand grant</a>                                                                   |
| 2025      | <a href="#">Recipient of the Ericsson Research Foundation grant</a>                                                                                                     |
| 2025      | Recipient of the 2025 American Control Conference (ACC) student travel award                                                                                            |
| 2025      | <a href="#">Recipient of WASP Research Stint Abroad</a>                                                                                                                 |
| 2024      | Recipient of Landshövding Per Westlings Minnesfond grant                                                                                                                |
| 2024      | Recipient of Sigfrid och Walborg Nordkvist grant                                                                                                                        |
| 2016-2021 | <a href="#">Recipient of Turkish government graduate program scholarship</a><br>A master program Scholarship to cover tuition fees, accommodation and a monthly stipend |

## Publications

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### Journal articles

- J1. **Bencherki, Fethi**, Türkay, S. & Akçay, H. Multi-input/multi-output switched-linear system identification from input-output data. *Signal Processing* (2025).
- J2. **Bencherki, Fethi**, Türkay, S. & Akçay, H. Observer-based switched-linear system identification. *Nonlinear Analysis: Hybrid Systems* **58**, 101620 (2025).
- J3. **Bencherki, Fethi** & Rantzer, A. Data-Driven Adaptive Dispatching Policies for Processing Networks. *IEEE Control Systems Letters* (2024).
- J4. **Bencherki, Fethi**, Türkay, S. & Akçay, H. Basis transform in linear switched system models from input–output data. *International Journal of Adaptive Control and Signal Processing* **37**, 3151–3168 (2023).
- J5. **Bencherki, Fethi**, Türkay, S. & Akçay, H. Realization of multi-input/multi-output switched linear systems from Markov parameters. *Nonlinear Analysis: Hybrid Systems* **48**, 101311 (2023).

### Peer-reviewed conference proceedings

- C1. **Bencherki, Fethi** & Rantzer, A. *Adaptive Control of Positive Systems with Application to Learning SSP* in *Proceedings of the 7th Annual Learning for Dynamics & Control Conference* **283** (PMLR, Apr. 2025), 660–672.
- C2. **Bencherki, Fethi**, Türkay, S. & Akçay, H. *MIMO-SLS Identification from Input-Output Data* in *2024 IEEE 18th International Conference on Control & Automation (ICCA)* (2024), 62–65.
- C3. **Bencherki, Fethi**, Türkay, S. & Akçay, H. *Realization of MIMO-SLSs from Markov Parameters via Forward/Backward Corrections* in *2024 European Control Conference (ECC)* (2024), 3138–3143.
- C4. **Bencherki, Fethi** & Rantzer, A. *Robust simultaneous stabilization via minimax adaptive control* in *2023 62nd IEEE Conference on Decision and Control (CDC)* (2023), 2503–2508.
- C5. Ohlin, D., **Bencherki, Fethi** & Tegling, E. *Achieving consensus in networks of increasingly stubborn voters* in *2022 IEEE 61st Conference on Decision and Control (CDC)* (2022), 3531–3537.

## Extended abstracts

- A1. **Bencherki, Fethi** & Rantzer, A. *Learning Optimal Queue Dispatch Policies* in *Reglermöte* (2025).
- A2. **Bencherki, Fethi** & Rantzer, A. *Learning Optimal Dispatching Policies for Queueing Systems* in *26th International Symposium on Mathematical Theory of Networks and Systems, MTNS* (2024).

## Preprints

- P1. **Bencherki, Fethi** & Akçay, H. *Asymptotic error bounds in observer-based system identification* 2025.

## Presentations

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### Talks

- T1. **Bencherki, Fethi**. *Adaptive control in the presence of uncertainty* Presented my work at Department of Electrical and Systems Engineering, the University of Pennsylvania (Philadelphia, PA, US). Apr. 2025.
- T2. **Bencherki, Fethi**. *Adaptive Control of Positive Systems* Licentiate Thesis Presentation (Lund, Sweden). [\[slides\]](#). Mar. 2025.
- T3. **Bencherki, Fethi**. *Data-Driven Adaptive Dispatching Policies for Processing Networks* American Control Conference (ACC25) (Denver, CO, US). [\[slides\]](#). July 2025.
- T4. **Bencherki, Fethi**. *Data-driven adaptive dispatching policies for queueing systems* 26th International Symposium on Mathematical Theory of Networks and Systems (Cambridge, UK). [\[slides\]](#). Aug. 2024.
- T5. **Bencherki, Fethi**. *MIMO-SLS identification from input-output data* 18th IEEE International Conference on Control & Automation (ICCA 2024) ( Reykjavik, Iceland). [\[slides\]](#). June 2024.
- T6. **Bencherki, Fethi**. *Realization of MIMO-SLSs from Markov parameters via forward/backward corrections* European Control Conference 2024 (ECC24) (Stockholm, Sweden). [\[slides\]](#). June 2024.
- T7. **Bencherki, Fethi**. *Robust Simultaneous Stabilization via Minimax Adaptive Control* 62nd IEEE Conference on Decision and Control (CDC24) (Singapore). [\[slides\]](#). Dec. 2023.

### Posters

- P1. **Bencherki, Fethi** & Rantzer, A. *Adaptive Control of Positive Systems with Application to Learning SSP* 7th Annual Learning for Dynamics & Control Conference (L4DC) (University of Michigan, Ann Arbor, US). [\[Poster\]](#). June 2025.
- P2. **Bencherki, Fethi** & Rantzer, A. *Learning Optimal Dispatching Policies in Queueing Systems* WASP winter conference (Norrköping, Sweden). [\[Poster\]](#). Jan. 2024.
- P3. **Bencherki, Fethi** & Rantzer, A. *Efficient Adaptive Robust Stabilization* WASP winter conference (Norrköping, Sweden). [\[Poster\]](#). Jan. 2023.
- P4. **Bencherki, Fethi** & Rantzer, A. *Guarantees on multiple model robust stabilization via Minimax adaptive control* WASP winter conference (Norrköping, Sweden). [\[Poster\]](#). Jan. 2022.
- P5. Ohlin, D., **Bencherki, Fethi** & Tegling, E. *Achieving consensus in networks of increasingly stubborn voters* ELLIIT Annual Workshop (linköping, Sweden). [\[Poster\]](#). Oct. 2021.

## Teaching

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### Lund University

2021-2024      Teaching Assistant and Lab Responsible in *Non-Linear Control and Servo Systems course*  
2021-2024      Teaching Assistant and Lab Responsible in *Automatic control, basic course*

## Academic service

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### Journal reviewer

2024              [Nonlinear Analysis: Hybrid Systems](#)  
2024              [IEEE Control Systems Letters \(L-CSS\)](#)  
2025              [IEEE Transactions on Automatic Control](#)  
2025              [Automatica](#)

### Conference reviewer

2025              [American Control Conference](#), Denver, Colorado, US, 8-10 July, 2025  
2024              [IEEE Conference on Decision and Control](#), Milan, Italy, December 16-19, 2024  
2024              [European Control Conference](#), Stockholm, Sweden, 25-28 June, 2024  
2024              [7th Annual Learning for Dynamics & Control Conference](#), Michigan, Ann Arbor, 4-6 June, 2025

### Organized conferences

2024              [European Control Conference 2024 \(ECC24\)](#), KTH Royal Institute of Technology, Stockholm, Sweden, 25-28 June, 2024.  
Role: Volunteer to help with registration, parallel sessions, etc

## Skills

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### Programming languages

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|---------------------|-----------------------|
| Competent           | Python                |
| Previous experience | C++, R, Matlab, Julia |
| Tools               | Git, $\LaTeX$         |

## References

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- [Prof. Anders Rantzer](#), Professor of Automatic Control at [Lund University](#), Dept.of Automatic Control, Lund, Sweden, email: [anders.rantzer@control.lth.se](mailto:anders.rantzer@control.lth.se)
- [Prof. Hüseyin Akçay](#), Professor of Automatic Control at [Eskisehir Technical University](#), Dept.of Electrical and Electronics Engineering, Eskisehir, Turkey, email: [huakcay@eskisehir.edu.tr](mailto:huakcay@eskisehir.edu.tr)